Ranking, Calibration and Cost in Competing-Risks Survival Models: A Common-Protocol Benchmark on Mortgage Data

Amirreza Sadraei, Ali Emre Nebiler, Filip Becic
Department of Computer and Systems Sciences, Stockholm University

***Abstract*—Lenders must state how likely a loan is to default by a given date, not merely which loans are riskiest, and a loan can end by default, prepayment or maturity. Deep competing-risks models are usually reported on ranking alone. We benchmark cause-specific Cox, Fine–Gray, a random survival forest and DeepHit on 550,000 Freddie Mac mortgages under one protocol that measures discrimination, calibration and computational cost together, with vintage-ordered splits and a tuning budget fixed in advance at twenty configurations for DeepHit and eight for the forest. DeepHit's advantage depends on the regime: with tuning it beats a cause-specific Cox model by 0.019 AUC on crisis-era loans and loses to it by 0.009 on recent ones. Its calibration penalty does not reverse, and it never beats a random survival forest. A pooled view of 40 tuning configurations appears to show that DeepHit's ranking-loss weight strongly affects both discrimination and calibration; a controlled ablation holding architecture fixed shows the effect is small and reverses sign across cohorts and causes. We also pre-registered and failed to confirm the hypothesis that selecting on discrimination costs calibration.**

I. Introduction

Credit scoring has long been posed as classification at a fixed horizon: will this borrower default within a year? Provisioning rules ask something harder. A lender must estimate losses over the remaining life of a loan, which requires a probability of default at every future date, and loans end in more than one way. A mortgage may default, be repaid early, or run to maturity, and once one of these happens the others cannot. Under such competing risks the quantity needed is the cumulative incidence function of default rather than one minus a survival curve [1]. The distinction is not academic: treating early repayment as missing information biases the estimated default probability upward [2], and a model can order borrowers correctly while attaching the wrong numbers to them.

Deep models that predict a cumulative incidence per cause exist and have been applied to mortgages [3], [4], but they are selected and reported on ranking. Whether their reported advantage survives when the probabilities themselves are measured is not established. This paper tests that question under a single protocol. Our contributions are: (i) a protocol that measures discrimination, calibration and computational cost together, on vintage-ordered splits, with a tuning budget fixed in advance and reported; (ii) a benchmark of four competing-risks models across two economic regimes on 550,000 mortgages, with paired significance tests and a seed study; and (iii) two negative results, one about DeepHit's ranking loss and one about the effect of the selection criterion.

II. Background and Related Work

Two recent reviews bracket this problem without meeting in the middle. Shang et al. [5] survey three decades of survival analysis in credit risk but do not examine how the resulting predictions are checked. Wiegrebe et al. [6] review more than fifty deep learning methods for time-to-event data and observe that most handle only a single risk with censoring; they consider no credit application. Neither asks whether credit models produce faithful probabilities.

On the credit side, the field's benchmark compares eight survival models on ten portfolios and finds that seven of them censor early repayment; on the same loans that choice moves the censoring rate from 20–85% to 94.56–98.16% [7]. Later work adopts competing risks explicitly, with cause-specific Cox models [8], random survival forests [2] and boosting [9], while macroeconomic covariates have long been used in the single-risk setting [10]. Blumenstock et al. [4] benchmarked DeepHit and a random survival forest against cause-specific Cox and Fine–Gray on US mortgages with default and prepayment as competing risks, and reported the highest concordance index for DeepHit in every experiment, evaluated over ten random train/test splits. Our model set and event pair are theirs; what we change is the protocol, and the present paper is therefore a replication under vintage-ordered splits with calibration

and cost added rather than a new benchmark design. On the evaluation side, DeepHit's loss combines a likelihood with a ranking term whose weight is tuned on validation discrimination [3]; Haider et al. [11] define D-calibration for single events, and Sonabend et al. [12] show that a concordance index computed on a predicted distribution depends on an unstated reduction to a single risk. Most recently, Alberge et al. [13] show that existing calibration measures are unsuited to competing risks, introduce two proper measures, and give recalibration methods that improve probabilities while preserving discrimination. These measures have not been applied to credit data, and no study evaluates ranking, calibration and cost for these models together.

Two qualifications matter for how that gap should be read. First, the deployed baseline is not a survival model: reviewing bank disclosures, Bank and Eder [14] find that lifetime probabilities of default under IFRS 9 are built mainly from transformations of regulatory estimates and rating-based matrix models, and that where competing risks are modelled at all the usual device is a discrete-time multinomial logit rather than a Fine–Gray or deep model. Competing-risks treatment of prepayment has therefore reached production, but in a model family absent from the benchmarking literature. Second, calibration is not neglected in practice. Supervisory guidance defines calibration as matching a long-run average default rate, and current work in the IFRS 9 term-structure setting still censors competing risks for tractability: Botha and Verster [15] report that 47% of the mortgage spells in their study end in a competing risk, 96% of them early settlements, and treat those spells as right-censored. The gap is thus a mismatch between the quantity the benchmarking literature reports and the quantity deployment requires, not an absence of concern for probabilities. Wycinka's comparison of competing-risks specifications on consumer loans points the same way: discrimination separated the models far less than their agreement with the empirical cumulative incidence did [16].

III. Protocol

A. Data

Our data are 50,000-loan samples per origination year from the Freddie Mac Single-Family Loan-Level Dataset: 300,000 loans originated 2013–2018 and 250,000 originated 2005–2009, with performance observed to March 2026 (Table I). Default is the first month at 90 or more days past due, following [4]; prepayment is zero-balance code 01. Time is loan age in months, so the origin is origination. These data reproduce the contrast

TABLE I
THE TWO COHORTS. "CENSORING, DEFAULT ONLY" RECOMPUTES THE CENSORING RATE WHEN PREPAYMENT IS TREATED AS CENSORED RATHER THAN AS AN EVENT.

Cohort	n	Default	Prepaid	Cens.	Cens. (def.)
Post-crisis	300,000	4.5%	75.0%	20.6%	95.5%
Crisis	250,000	10.5%	87.1%	2.4%	89.5%

that motivates the competing-risks framing: counting prepayment as an event leaves 20.6% of loans censored, while censoring it leaves 95.5%. Dirick et al. [7] report 20–85% and 94.56–98.16% on ten European consumer-credit portfolios; ours sit at the lower edge of the first range and inside the second, on a different product, country and decade. Prepayment is not a nuisance here but the dominant outcome, outnumbering default seventeen to one.

B. Models and the common interface

Every model implements the same interface, taking covariates, times and event labels and returning a predicted cumulative incidence per cause on a fixed horizon grid, so no model can be given a metric the others cannot. We compare the Aalen–Johansen estimator as a null model that cannot discriminate but is calibrated by construction, a cause-specific Cox model, a Fine–Gray subdistribution model, a random survival forest and DeepHit. Two implementation details matter for interpretation. Cox and Fine–Gray were implemented directly, since no mainstream Python library provides Fine–Gray; both maximise a Breslow partial likelihood, with Fine–Gray retaining subjects who experience the competing event in the risk set, weighted by the ratio of the censoring survival function at the current time to its value at their event time, as in [1]. Our forest is a cause-specific random survival forest, fitting one forest per cause, and is therefore a relative of rather than a replication of the composite-splitting competing-risks forest used by Frydman and Matuszyk [2].

C. Metrics

Discrimination is the inverse-probability-of-censoring-weighted time-dependent AUC at 12 to 60 months, with subjects who experience the competing event treated as controls because they can never experience the cause of interest. Calibration is measured in two ways: the integrated Brier score for the cumulative incidence, and a calibration index comparing mean predicted risk against the Aalen–Johansen estimate within deciles of predicted risk. It is a binned form of the integrated calibration

index [17], whose competing-risks version smooths the calibration curve rather than binning it [18]. Cost is the measured fit and prediction time with the hardware recorded. Censoring weights come from a Kaplan–Meier estimate fitted on the training set only. Because the time origin is origination and every vintage has at least eight years of follow-up, censoring before 60 months is rare: $G(60) = 0.9986$ in both cohorts, so these weights are near one and it is the competing event, not censoring, that shapes the results. We do not implement the per-cause calibration measures of Alberge et al. [13]; adding them is the clearest extension of this work.

D. Splits, tuning budget and hypotheses

Models are trained on earlier origination years and tested on later ones, which is how a lender uses them, in contrast to the random splits that dominate the literature [4]. The post-crisis cohort trains on 2013–2015 and tests on 2016–2018; the crisis cohort trains on 2005–2006 and tests on 2007–2009. Hyperparameters were searched with a budget fixed in advance: twenty random configurations for DeepHit and eight for the forest, scored on an out-of-time validation vintage held inside the training years, so test vintages were never touched during tuning. Cox and Fine–Gray have no comparable hyperparameters and are reported untuned. Each configuration was scored on validation AUC and on validation integrated Brier score, and the winner under each criterion was refitted and evaluated separately.

For inference we fixed, before examining any test result, a two-sided null hypothesis that the models do not differ, a significance level of 0.05, and a Bonferroni correction across the three metric families, giving a threshold of 0.0167. A family is the three metrics of one pre-specified comparison in one cohort; thirty-six tests are computed in all, so the correction holds within a family, not across the table. Differences are assessed by a paired bootstrap over the test set with 500 resamples, which cannot resolve p below $1/500$, so 0.002 is reported as an upper bound. Because the bootstrap holds the fitted model fixed and therefore understates total uncertainty, DeepHit was additionally refitted with five seeds per cohort; its seed-to-seed standard deviation was 0.0014 and 0.0010 in AUC, and we treat differences below twice that as indistinguishable regardless of their p-value.

IV. Results

A. The advantage reverses with regime

DeepHit's standing depends on which economic regime it is tested in (Table II). On the crisis cohort the tuned model reaches an AUC of 0.8538 against 0.8346 for the cause-specific Cox model, a difference of $+0.0192$ that is significant under the paired bootstrap. That margin should be read with care: refitting on bootstrap resamples of the training vintages (Table V) gives the Cox model a standard deviation of 0.0159 in this cohort, against 0.0035 for DeepHit, so the gap is little more than one standard deviation of the baseline's own training variability: here the classical models are the unstable ones. On the post-crisis cohort the ordering reverses: 0.7280 against 0.7372, a difference of -0.0091 that is also significant (Table IV) and, with every training standard deviation below 0.0022 in that cohort, comfortably outside the noise. The calibration comparison does not reverse. In both cohorts DeepHit's calibration index is 0.0040 worse than Cox's, so the regime that decides whether the deep model ranks better does not change the direction of the calibration gap (Fig. 1, left). The two cohorts carry it with very different force: post-crisis the gap is about eighteen times the combined training variability of the two models, on crisis data only about 1.7 times it. The sign replicates; the size does not. Every curve in that figure lies above the diagonal, so calibration here is limited less by model class than by the vintage shift itself.

TABLE II
Discrimination (AUC at 60 months) and calibration (decile-binned ICI at 60 months) for default, on the held-out test vintages of each cohort.

	Crisis (2007–09)		Post-crisis (2016–18)	
Model	AUC	ICI	AUC	ICI
Aalen–Johansen	0.5000	0.0077	–	–
Cox	0.8346	0.0169	0.7372	0.0278
Fine–Gray	0.8327	0.0159	–	–
RSF	0.8516	0.0286	0.7300	0.0287
RSF (tuned)	0.8538	0.0267	0.7275	0.0289
DeepHit α=0.2	0.8301	0.0281	0.7239	0.0242
DeepHit α=1	0.8405	0.0240	0.7246	0.0318
DeepHit (tuned)	0.8538	0.0209	0.7280	0.0318

B. Nothing is gained over a random survival forest

Against a tuned random survival forest, DeepHit offers nothing. The two are statistically indistinguishable on discrimination in both cohorts, with $p = 0.968$ on crisis data and $p = 0.788$ post-crisis, and both differences fall inside DeepHit's own seed variation. These are among the few comparisons in this study where we fail to reject: whatever the deep model captures, a forest captures as well. On prepayment in the crisis cohort the forest is distinctly better, reaching an AUC of 0.685 against 0.620 and a calibration index of 0.154 against 0.236. The

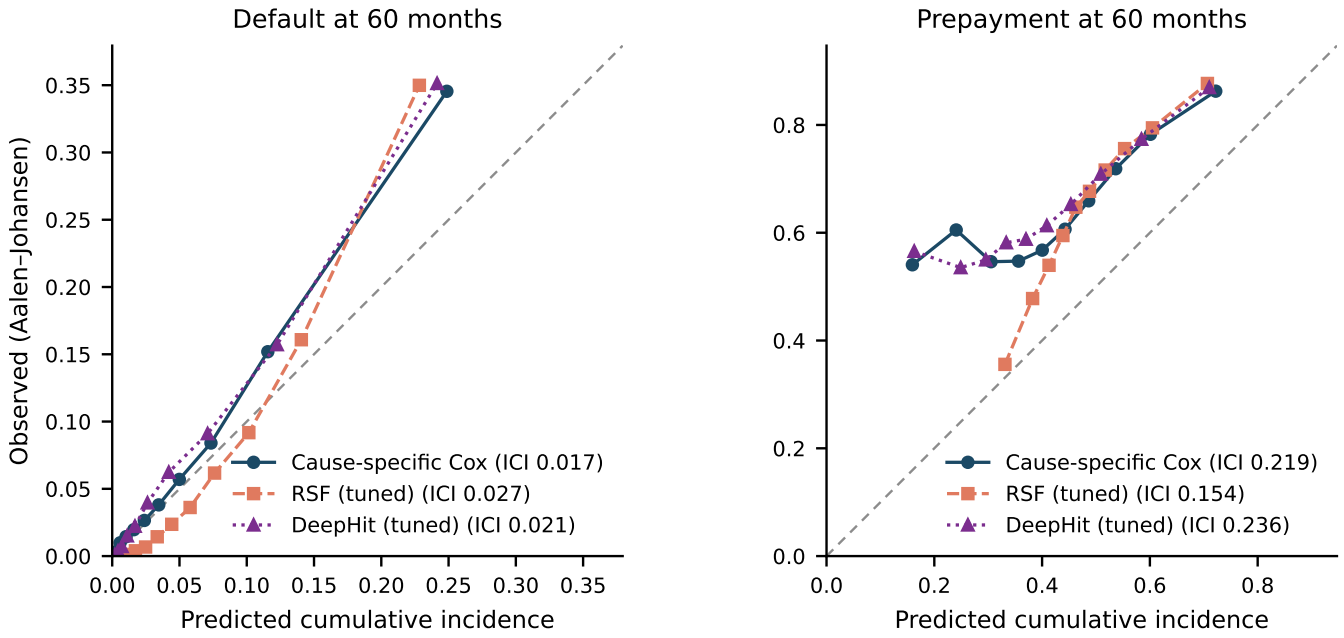


Fig. 1. Calibration at 60 months on the crisis cohort (train 2005–06, test 2007–09). Each point is a decile of predicted risk; the vertical axis is the Aalen–Johansen estimate within that decile, so competing events and censoring are both accounted for. A model on the diagonal is perfectly calibrated; every curve lies above it, so all three under-predict out of time. The deciles separate by a few percentage points for default and by twenty or more for prepayment, the competing event the credit literature most often censors.

competing event, which the credit literature most often censors, is where the gap between the two is widest (Fig. 1, right).

C. The loss weight is not a reliable knob

DeepHit's loss is a convex combination α·likelihood $+(1-\alpha)$·ranking, and the original model selects this weight on validation discrimination [3]. Pooling our forty tuning configurations by α suggests a strong monotone effect, mean validation AUC rising from 0.658 at $\alpha = 0.1$ to 0.776 at $\alpha = 1.0$, but that is a marginal over a search in which α varied alongside depth, dropout, learning rate, batch size and bin count, and it does not survive control.

Holding everything else fixed and varying only α, over five seeds and both cohorts, the effect is small and inconsistent in sign (Table III). On post-crisis default it is invisible in discrimination, $+0.0005$ against a seed standard deviation of 0.0018, while calibration worsens steadily as the ranking term is removed, from 0.0189 to 0.0303. On crisis prepayment the pattern inverts: removing the ranking term costs 0.067 of AUC and worsens calibration. On post-crisis prepayment, removing it *improves* both. Of the four cause-by-cohort cells, three show a discrimination effect larger than seed noise and they do not agree on direction.

The loss weight is therefore not the dominant hyperparameter the pooled view suggested, and no setting of it can be recommended from these data. The discrepancy between the two views is itself a result: a marginal taken over a random search is not an ablation, and reading one as the other is how a hyperparameter acquires a reputation it has not earned.

TABLE III
CONTROLLED ABLATION OF DEEPHIT'S LOSS WEIGHT α: ARCHITECTURE, LEARNING RATE, BATCH SIZE, DROPOUT AND BIN COUNT HELD FIXED, FIVE SEEDS PER SETTING, EVALUATED AT 60 MONTHS ON THE HELD-OUT TEST VINTAGES. $\alpha = 1$ REMOVES THE RANKING TERM ENTIRELY. THE EFFECT REVERSES SIGN ACROSS COHORTS AND CAUSES, AND IN ONE CELL IS SMALLER THAN THE SEED STANDARD DEVIATION. Δ IS THE CHANGE FROM α=0.1 TO α=1.

Cause	Cohort	ΔAUC	seed sd	ΔICI
def.	crisis	+0.0075	0.0033	-0.0010
def.	post	+0.0005	0.0018	+0.0114
prep.	crisis	-0.0673	0.0103	+0.0345
prep.	post	+0.0386	0.0062	-0.0397

D. Selecting on discrimination does not cost calibration

We pre-registered the hypothesis that choosing a configuration on validation discrimination would yield a worse calibrated model than choosing on a probability-accuracy criterion, and we did not confirm it. Of four model-by-cohort comparisons, three produce differences

TABLE IV
PAIRED BOOTSTRAP ON THE TUNED MODELS (B=500), DEFAULT, 60 MONTHS. T1 DEEPHIT VS COX; T2 DEEPHIT VS RSF; T3 DEEPHIT SELECTED ON AUC VS ON IBS. * SIGNIFICANT AT THE BONFERRONI LEVEL $0.05/3$; † DIFFERENCE SMALLER THAN TWICE DEEPHIT'S SEED-TO-SEED SD.

Test	Cohort	Metric	Diff	95% CI	p	
T1	post	AUC	-0.0091	[-0.0116, -0.0066]	<0.002	*
T1	post	ICI	+0.0040	[+0.0040, +0.0041]	<0.002	*
T2	post	AUC	+0.0005	[-0.0024, +0.0038]	0.788	†
T2	post	ICI	+0.0030	[+0.0029, +0.0030]	<0.002	*
T3	post	AUC	+0.0005	[-0.0017, +0.0026]	0.604	†
T3	post	ICI	+0.0015	[+0.0014, +0.0015]	<0.002	*
T1	crisis	AUC	+0.0192	[+0.0167, +0.0214]	<0.002	*
T1	crisis	ICI	+0.0040	[+0.0038, +0.0043]	<0.002	*
T2	crisis	AUC	-0.0000	[-0.0019, +0.0020]	0.968	†
T2	crisis	ICI	-0.0057	[-0.0069, -0.0041]	<0.002	*
T3	crisis	AUC	+0.0101	[+0.0091, +0.0112]	<0.002	*
T3	crisis	ICI	+0.0019	[+0.0017, +0.0022]	<0.002	†

TABLE V
TRAINING VARIABILITY ON THE **CRISIS** COHORT: EACH MODEL REFITTED ON FIVE BOOTSTRAP RESAMPLES OF ITS TRAINING VINTAGES AND EVALUATED ON THE FIXED TEST VINTAGES (DEFAULT, 60 MONTHS). SEEDS ARE THE WRONG DEVICE FOR COX AND FINE–GRAY, WHICH ARE DETERMINISTIC GIVEN THE DATA, SO THE TRAINING SET IS RESAMPLED INSTEAD. EVERY POST-CRISIS STANDARD DEVIATION IS BELOW 0.0022 AND IS OMITTED.

Model	AUC mean	sd	ICI mean	sd
Cox	0.8167	0.0159	0.0188	0.0020
Fine–Gray	0.8316	0.0206	0.0202	0.0024
RSF	0.8513	0.0009	0.0255	0.0005
DeepHit	0.8502	0.0035	0.0211	0.0012

smaller than twice DeepHit's seed-to-seed variation, including every post-crisis comparison. The one robust effect is a $+0.0101$ AUC gain in the crisis cohort for the configuration chosen on AUC, which is selection behaving as intended. The selection criterion does not drive the calibration gap.

E. Recalibration does not help, and why

Since Alberge et al. [13] show that recalibration improves competing-risks probabilities while preserving discrimination, we applied a marginal per-horizon rescaling in that spirit, estimating it two ways (Table VI). Fitted on the training vintages, which is all a lender has when scoring a new loan, it changes almost nothing: in ten of sixteen cells it leaves the calibration index worse than the raw predictions. Fitted instead on the test vintages — an oracle no lender could build — it reduces the index by a median factor of 4.2, reaching 14.7 for Fine–Gray on post-crisis default. The single exception is the forest on crisis default, which the oracle leaves unchanged.

TABLE VI
POST-HOC RECALIBRATION, ICI AT 60 MONTHS. "TRAIN" ESTIMATES THE RESCALING ON THE TRAINING VINTAGES, WHICH IS ALL A LENDER HAS AT SCORING TIME; "ORACLE" ESTIMATES IT ON THE TEST VINTAGES AND IS UNATTAINABLE IN PRACTICE. COX AND DEEPHIT ARE SHOWN; THE FOREST AND FINE–GRAY BEHAVE THE SAME WAY. AUC IS IDENTICAL TO FOUR DECIMALS FOR RAW AND RECALIBRATED PREDICTIONS IN ALL SIXTEEN CELLS, CONFIRMING THE RESCALING IS MONOTONE.

Cohort	Cause	Model	Raw	Recal. train	Recal. oracle
crisis	def.	Cox	0.0169	0.0182	0.0061
crisis	def.	DeepHit	0.0209	0.0164	0.0053
crisis	prep.	Cox	0.2186	0.2204	0.1168
crisis	prep.	DeepHit	0.2361	0.2222	0.1063
post	def.	Cox	0.0278	0.0279	0.0025
post	def.	DeepHit	0.0318	0.0297	0.0082
post	prep.	Cox	0.1640	0.1693	0.0253
post	prep.	DeepHit	0.1850	0.1685	0.0439

The contrast identifies what the calibration penalty actually is. It is not a defective calibration map, or the oracle correction would not work so well. It is the vintage shift itself: a rescaling estimated on 2005–06 cannot know that 2007–09 would turn out worse. Recalibration as demonstrated in the literature is an in-distribution procedure, and extending it across origination vintages is an open problem rather than a step we omitted.

F. What the evaluation protocol is worth

The benchmarking literature evaluates on random splits; we evaluate on vintage-ordered ones. Holding the models and their tuned settings fixed and changing only how the test set is drawn isolates what that choice is worth (Table VII). Under random splits DeepHit leads both baselines in all eight cause-by-cohort cells, by between $+0.0022$ and $+0.0178$ AUC, and the estimates are stable: no seed standard deviation exceeds 0.0019. Under vintage ordering three of those eight leads reverse. DeepHit falls behind the Cox model on post-crisis default, draws level with the forest on crisis default, and loses to it on crisis prepayment by 0.065.

The mechanism is not only leakage. A random split both places temporal neighbours of each test loan in the training set and removes the shift between origination vintages, and a lender scoring a new loan faces neither advantage. The two effects cannot be separated here, and absolute scores do not move in one direction: on post-crisis default every model gains between $+0.021$ and $+0.033$ AUC from random splitting, while on crisis default the two flexible models lose, because the 2007–09 test vintages carry more default signal than a random

TABLE VII
WHAT THE EVALUATION PROTOCOL IS WORTH. DEEPHIT'S AUC ADVANTAGE AT 60 MONTHS OVER THE CAUSE-SPECIFIC COX MODEL AND OVER THE TUNED FOREST, MEASURED UNDER RANDOM SPLITS (MEAN OF FIVE SEEDS, MATCHED TO THE VINTAGE SPLIT'S TRAIN AND TEST SIZES; EVERY SEED STANDARD DEVIATION IS AT OR BELOW 0.0019) AND UNDER VINTAGE-ORDERED SPLITS. MODELS AND TUNED SETTINGS ARE IDENTICAL THROUGHOUT; ONLY THE WAY THE TEST SET IS CHOSEN CHANGES. UNDER RANDOM SPLITS DEEPHIT LEADS IN ALL EIGHT CELLS; UNDER VINTAGE ORDERING THREE OF THOSE LEADS REVERSE.

		vs Cox		vs RSF	
Cause	Cohort	random	vintage	random	vintage
def.	crisis	+0.0025	+0.0192	+0.0032	-0.0000
def.	post	+0.0022	-0.0091	+0.0035	+0.0005
prep.	crisis	+0.0178	+0.0012	+0.0104	-0.0651
prep.	post	+0.0139	+0.0012	+0.0108	+0.0048

draw does. What moves consistently is the comparison between models, and it moves in DeepHit's favour. Since the tuned configurations were selected on an out-of-time inner validation, they are if anything suited to the vintage protocol, so the gap we measure is a lower bound on the protocol's effect.

G. Cost

The models span three orders of magnitude in training cost. On 150,000 loans, Fine–Gray fits in 0.03 seconds and the cause-specific Cox model in 0.42, against 36 seconds for the forest and 59 for DeepHit, and the DeepHit search added 14.4 minutes of fitting across the two cohorts and the forest search 4.1, while the classical models have no search to run. Since the deep model is indistinguishable from the forest and loses to Cox in one of the two regimes, that factor of roughly two thousand buys no reliable improvement. We report these figures because the benchmarking literature generally omits them [6].

V. DISCUSSION, LIMITATIONS AND CONCLUSION

Taken together these results qualify, rather than contradict, the finding that deep competing-risks models perform well on mortgage data [4]. DeepHit does win on crisis-era loans once tuned; an untuned deep baseline is not evidence. But the win is confined to one regime, disappears against a random survival forest, costs three orders of magnitude more compute, and is accompanied by a calibration penalty in every condition we tested. For a lender who needs a probability rather than an ordering, that combination is difficult to justify.

Several limitations bound these conclusions. Our recalibration is a marginal per-horizon rescaling rather than the per-cause procedures of Alberge et al. [13]; a stronger correction might close more of the gap, though the oracle column suggests the ceiling is set by what is knowable at scoring time rather than by the form of the map. Training variability is estimated from five bootstrap resamples, which conflates sampling with optimisation noise. Our covariates are observed at origination, with no time-varying macroeconomic variables of the kind introduced by Bellotti and Crook [10]. The forest is cause-specific rather than Ishwaran's competing-risks formulation, and all results come from a single family of US agency mortgages. Finally, with 150,000 test subjects almost any difference attains significance, which is why we report effect sizes against seed variation throughout.

Under one protocol measuring ranking, calibration and cost together on vintage-ordered splits, deep competing-risks models offer no reliable advantage over classical or tree-based baselines on mortgage data, are worse calibrated in both regimes, and depend on the regime for whatever ranking advantage they show. The result that generalises past this benchmark is methodological: a hyperparameter's apparent effect in a pooled search did not survive controlled variation.

CODE AND DATA

Pipeline, models, metrics and experiments: https://github.com/ar-sadraei/rtds-project, which rebuilds every table in this paper. The Freddie Mac data require registration and cannot be redistributed.

REFERENCES

[1] J. P. Fine and R. J. Gray, "A proportional hazards model for the subdistribution of a competing risk," *Journal of the American Statistical Association*, vol. 94, no. 446, pp. 496–509, 1999.

[2] H. Frydman and A. Matuszyk, "Random survival forest for competing credit risks," *J. Oper. Res. Soc.*, vol. 73, no. 1, pp. 15–25, 2022. [Online]. Available: https://doi.org/10.1080/01605682.2020.1759385

[3] C. Lee, W. R. Zame, J. Yoon, and M. van der Schaar, "Deephit: A deep learning approach to survival analysis with competing risks," in *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, (AAAI-18), the 30th innovative Applications of Artificial Intelligence (IAAI-18), and the 8th AAAI Symposium on Educational Advances in Artificial Intelligence (EAAI-18), New Orleans, Louisiana, USA, February 2-7, 2018*, S. A. McIlraith and K. Q. Weinberger, Eds. AAAI Press, 2018, pp. 2314–2321. [Online]. Available: https://doi.org/10.1609/aaai.v32i1.11842

[4] G. Blumenstock, S. Lessmann, and H. Seow, "Deep learning for survival and competing risk modelling," *J. Oper. Res. Soc.*, vol. 73, no. 1, pp. 26–38, 2022. [Online]. Available: https://doi.org/10.1080/01605682.2020.1838960

[5] L. Shang, J. Zhao, G. Li, and X. Zhang, "Survival analysis in credit risk management: a review study," *The Journal of Credit Risk*, vol. 20, no. 4, 2024.

[6] S. Wiegrebe, P. Kopper, R. Sonabend, B. Bischl, and A. Bender, "Deep learning for survival analysis: a review," *Artificial Intelligence Review*, vol. 57, no. 3, p. 65, 2024.

[7] L. Dirick, G. Claeskens, and B. Baesens, "Time to default in credit scoring using survival analysis: a benchmark study," *J. Oper. Res. Soc.*, vol. 68, no. 6, pp. 652–665, 2017. [Online]. Available: https://doi.org/10.1057/s41274-016-0128-9

[8] M. Thackham and J. Ma, "On maximum likelihood estimation of competing risks using the cause-specific semi-parametric cox model with time-varying covariates - an application to credit risk," *J. Oper. Res. Soc.*, vol. 73, no. 1, pp. 5–14, 2022. [Online]. Available: https://doi.org/10.1080/01605682.2020.1800418

[9] C. A. P. B. Saavedra, J. B. F. Gomes, E. M. de Castro Gomes, and H. Kimura, "Probability of default for lifetime credit loss for IFRS 9 using machine learning competing risks survival analysis models," *Expert Syst. Appl.*, vol. 249, p. 123607, 2024. [Online]. Available: https://doi.org/10.1016/j.eswa.2024.123607

[10] T. Bellotti and J. Crook, "Credit scoring with macroeconomic variables using survival analysis," *J. Oper. Res. Soc.*, vol. 60, no. 12, pp. 1699–1707, 2009. [Online]. Available: https://doi.org/10.1057/jors.2008.130

[11] H. Haider, B. Hoehn, S. Davis, and R. Greiner, "Effective ways to build and evaluate individual survival distributions," *J. Mach. Learn. Res.*, vol. 21, pp. 85:1–85:63, 2020. [Online]. Available: https://jmlr.org/papers/v21/18-772.html

[12] R. Sonabend, A. Bender, and S. J. Vollmer, "Avoiding c-hacking when evaluating survival distribution predictions with discrimination measures," *Bioinform.*, vol. 38, no. 17, pp. 4178–4184, 2022. [Online]. Available: https://doi.org/10.1093/bioinformatics/btac451

[13] J. Alberge, T. Haugomat, G. Varoquaux, and J. Abécassis, "On the calibration of survival models with competing risks," in *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, ser. Proceedings of Machine Learning Research, E. Khan, Y. Li, A. Solin, and A. Ramdas, Eds., vol. 300. PMLR, 2026, pp. 1837–1845. [Online]. Available: https://proceedings.mlr.press/v300/alberge26a.html

[14] M. Bank and B. Eder, "A review on the probability of default for IFRS 9," SSRN, Tech. Rep., 2021.

[15] A. Botha and T. Verster, "Approaches for modelling the term-structure of default risk under IFRS 9: a tutorial using discrete-time survival analysis," *International Journal of Data Science and Analytics*, vol. 22, no. 1, 2026.

[16] E. Wycinka, "Competing risk models of default in the presence of early repayments," *Econometrics*, vol. 23, no. 2, pp. 99–120, 2019.

[17] P. C. Austin and E. W. Steyerberg, "The integrated calibration index (ICI) and related metrics for quantifying the calibration of logistic regression models," *Statistics in Medicine*, vol. 38, no. 21, pp. 4051–4065, 2019.

[18] P. C. Austin, H. Putter, D. Giardiello, and D. van Klaveren, "Graphical calibration curves and the integrated calibration index (ICI) for competing risk models," *Diagnostic and Prognostic Research*, vol. 6, no. 1, p. 2, 2022.

[19] H. Kvamme, Ø. Borgan, and I. Scheel, "Time-to-event prediction with neural networks and cox regression," *J. Mach. Learn. Res.*, vol. 20, pp. 129:1–129:30, 2019. [Online]. Available: https://jmlr.org/papers/v20/18-424.html